Assignment 8: Hardware Setup and Simulation Documentation

Internship: IoT with AI

Week: 1

Topic: Introduction to IoT and Electronics Basics

1. Introduction

Hardware setup is an important part of IoT development. It involves connecting microcontrollers, sensors, electronic components and other devices to develop functional electronic systems.

In this assignment, Arduino Uno and different electronic components were used to perform practical experiments through the Tinkercad online simulation platform.

2. Hardware and Software Requirements

Hardware Components

The following components were used in the simulation experiments:

1. Arduino Uno R3.
2. Breadboard.
3. Red, yellow and green LEDs.
4. Push button.
5. 220Ω resistors.
6. 10kΩ resistor.
7. TMP36 temperature sensor.
8. Jumper wires.

Software and Simulation Tools

1. Tinkercad Circuits – For designing and simulating electronic circuits.

2. Arduino C/C++ – For writing microcontroller programs.

3. Serial Monitor – For displaying sensor readings.

3. Hardware Setup

The Arduino Uno was used as the main microcontroller for the practical experiments.

A breadboard was used to connect electronic components without soldering. LEDs, push buttons and sensors were connected to the appropriate Arduino pins using jumper wires.

Since physical hardware was not available, all circuit connections and experiments were performed using Tinkercad Circuits.

4. Experiments Performed

Experiment 1: Three-LED Blinking Circuit

Three LEDs (red, yellow and green) were connected to Arduino Uno using individual 220Ω resistors.

The Arduino program controlled the LEDs sequentially using digital output pins and time delays.

Result: The three LEDs glowed one after another in a continuous sequence.

Experiment 2: Push Button Controlled LED

A push button was connected to Arduino digital pin 2, and an LED was connected to digital pin 13 through a 220Ω resistor.

The Arduino program used the INPUT_PULLUP configuration to detect button presses.

Result: The LED turned ON when the push button was pressed and turned OFF when it was released.

Experiment 3: Temperature Sensor Interfacing

A TMP36 temperature sensor was connected to Arduino Uno.

The sensor output was connected to analog pin A0. Arduino read the analog voltage, calculated the temperature and displayed the readings through the Serial Monitor.

Result: Temperature readings were successfully displayed in degrees Celsius.

5. Testing and Verification

The experiments were tested using the Tinkercad simulation environment.

The following operations were verified:

1. Sequential operation of three LEDs.
2. Push button input detection.
3. LED ON/OFF control.
4. Temperature sensor readings.
5. Serial Monitor output.

The simulation environment helped verify the circuit connections and Arduino programs without requiring physical electronic components.

6. Advantages of Online Simulation

1. Allows circuit designing without physical components.
2. Helps identify connection errors.
3. Provides a safe environment for testing circuits.
4. Supports Arduino programming and simulation.

5. Reduces the cost of initial hardware experimentation.

7. Conclusion

The hardware setup and simulation activities provided practical understanding of Arduino Uno, breadboard connections, electronic components and sensor interfacing. Using Tinkercad Circuits, different experiments were designed, programmed and tested successfully. These activities helped develop basic knowledge of electronics, microcontroller programming and IoT system development.